

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

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Practical 5

- a) Write a python code to demonstrate importing pandas libraries and create data frame object.

Program:

```
import pandas as pd
data = {
    "name": ["Riya ", "Nikita", "Jasad"],
    "age": [19, 18, 21],
    "qualified": [True, True, False]
}
df = pd.DataFrame(data)
newdf = df.filter(items=["name", "age"])
print(newdf)
print("sapna_29")
```

Output:

```
   name  age
0  Riya   19
1 Nikita  18
2  Jasad  21
sapna_29
```

b) Write a python code to show statistical information on given data set.

Program:

```
import statistics
print("Mean:")
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
print("\nMedian:")
print(statistics.median([1, 3, 5, 7, 9, 11]))
print("\nMode:")
print(statistics.mode(['red', 'green', 'blue', 'red']))
print("\nStandard Deviation:")
print(statistics.stdev([1, 30, 50, 100]))
print("\nVariance:")
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print("Sapna_29")
```

Output:

```
--- RESTART: C:/Users/
```

```
Mean:
1.8666666666666667
```

```
Median:
6.0
```

```
Mode:
red
```

```
Standard Deviation:
41.67633221226007
```

```
Variance:
14
Sapna_29
```

c) Write a python code to create pandas series from dictionaries.

Program:

```
import pandas as pd
dictionary = {'DS':10, 'PDS':20, 'AM': 30}
series = pd.Series(dictionary)
print(series)
print("Sapna_29")
```

Output:

```
DS      10
PDS     20
AM      30
dtype: int64
Sapna_29
```

d) Write a python code to demonstrate filter pandas series with Boolean arrays.

Program:

```
import pandas as pd
dict = {'name':["Prerna", "Raj", "Madhuri", "Cheeku"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[60, 70, 90, 88]}
df = pd.DataFrame(dict, index = [0, 1, 2, 3])
print(df[[False, False, True, False]])
print("sapna_29")
```

Output:

```
   name degree score
2 Madhuri M.Tech   90
sapna_29
```